

You give them something to eat: Dietary Diversity, Coverage, and Catholic Aid in Rwanda

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Abstract

How much does a food-security experiment add to the choice between reaching more households and providing more to each? I reanalyze the 248-village evaluation of Catholic Relief Services' Gikuriro program in Rwanda. At its expected provider budget, a lower/large-cash lottery has the highest fitted mean diet, exceeding Gikuriro by 0.237 food groups among identified diets. This ranking does not establish a positive full-baseline advantage. I combine missing-diet intervals with coherent score distributions and the released assignment quotas. Support and costs alone permit a diversified loss bound of 9.600 groups; observed-outcome consistency lowers it to 8.074. A conditional exponential-mixture event gives a verified allocation bound of 5.868, compared with 7.720 from a looser finite construction. The new calculation retains a common exponential budget across packages and uses moment constants selected without follow-up outcomes. Its remaining width still precludes a small-loss recommendation. A much smaller block approximation fails synthetic coverage tests. Assistance requirements can increase regret against the original menu while lowering regret within a restricted menu. These results separate fitted effects, diversification and what the trial can defend, conditional on explicit assignment, weighting and delivery assumptions.

Keywords: food security; Catholic Relief Services; cash transfers; dietary diversity; treatment allocation.

JEL codes: C93, D61, I32, O12.

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1 Introduction

A food-security organization must decide both what to provide and how many households to reach. Giving a large transfer to a small share of villages can improve recipients' diets while leaving most villages untreated. A less expensive package can reach everyone and produce a smaller gain per recipient. An experimental comparison of package means informs this choice only after the untreated households, the provider budget, and uncertainty about the outcomes enter the same calculation.

I examine this question in the Gikuriro program in Rwanda. Catholic Relief Services (CRS) and SNV implemented the program with support from USAID. Nutrition and health activities, agricultural support, saving groups, and water, sanitation, and hygiene interventions formed a single package. Its stated aim was to improve maternal and child nutrition. The experiment of [McIntosh and Zeitlin \(2024\)](#) assigned 248 villages to Gikuriro, control, or one of four cash amounts delivered by GiveDirectly. The title recalls Mark 6:37. Religious affiliation was not randomized, so the experiment cannot identify a specifically Catholic delivery mechanism.

The parent study already develops equal-cost cash benchmarking, cost effectiveness, and the tradeoff between coverage and intensity. For mean diet, the Gikuriro comparison against a control/large-cash lottery is exactly its cost-effectiveness contrast multiplied by Gikuriro's cost. Its studentized statistic is unchanged. I therefore ask a narrower question: what allocation can be defended when the objective includes the diet distribution, missing outcomes are allowed to differ by assignment, and the organization can diversify across packages? The economic distinction is between the fitted winner and the opportunity cost that the evidence can credibly bound.

At the source's expected provider budget of \$124.49 per eligible household, large cash alone is unaffordable. A control/large lottery serves 24.1 percent of villages. A lower/large lottery serves every village with cash and reserves the large package for 15.3 percent. The latter has the highest fitted mean dietary diversity. Choosing Gikuriro instead has fitted loss 0.237 groups on the twelve-group household score. Missing-outcome endpoint means, however, allow the sign of this contrast to reverse. The stronger recipient-level response to large cash does not resolve the equal-budget decision.

The decision class itself supplies protection even before any outcomes are used. With only the twelve-group support and the source costs, the optimized worst-case mean loss is 9.600 groups. Assigning equal shares to the five individually affordable packages attains this bound. Its improvement over selecting a single package is diversification under ignorance, rather than information learned from the experiment. I report this benchmark before evaluating what the observed diets add.

Observed-outcome consistency tightens the bound to 8.074 groups for the fixed weighted released baseline sample. A finite conditional-assignment calculation adds simultaneous intervals for mean, survival and shortfall targets. Requiring those intervals to describe one score distribution per arm, and intersecting them with consistency, gives 7.720 groups. Relative to support alone, the total tightening is 1.880 groups, or 19.6 percent. Neither this conservative result nor the previously

looser mean box establishes that all valid analyses must be uninformative. The new projection demonstrates that the same outcomes and coverage event can support a stronger calculation.

Using the quota geometry more directly gives a verified mean-loss bound of 5.868 groups. I compute a conservative exponential-moment maximum over the bounded village populations using a sorted-threshold recurrence. A single sum-of-exponentials event then protects the common diet distributions without treating their uncertainty as six independent interval budgets. The resulting allocation spends \$84.27 per eligible household. Its bound is 1.852 groups below the earlier coherent construction. This comparison concerns two separately valid conditional procedures; it does not allocate a common coverage event to their intersection or select their smaller realized bound. The headline result uses the quota procedure alone. The calculation adapts established symmetric-convex and sampling arguments; no new general inference theorem or general range algorithm is claimed.

The much smaller exploratory block bound, 0.570 groups, remains unvalidated. Fixed quotas induce dependence across arms within 22 assignment blocks. Common block scores retain that covariance, but synthetic fifteen-comparison coverage is still 0.869 with equal weights and 0.740 with unequal frame weights. Those failures falsify a general nominal-coverage interpretation of the approximation; they do not estimate its unknown coverage for Rwanda. The finite event instead assumes exchangeable labels conditional on each released block’s counts, independent blocks and fixed baseline weights. It covers that released target, not unsampled households or deployment by itself.

The relevant opportunity cost also depends on the organization’s rules. Eliminating control raises the smallest mean-loss bound against the original menu from 7.720 to 8.138. Comparing only among universally assisted allocations produces a smaller regret number because the benchmark changes. Similarly, requiring a Gikuriro share can mechanically scale regret when it restricts both the choice and comparator. I distinguish these effects rather than treating a smaller within-menu bound as evidence that an assistance mandate improves diet. This is a change in the optimized worst-case bound, rather than a measured dietary opportunity cost. The rules are hypothetical institutional sensitivities, not observed CRS requirements.

The decision problem draws directly on [Manski \(2004, 2007\)](#) and [Stoye \(2007\)](#). Their work makes missing outcomes and multiple treatments part of treatment choice. [Kitagawa and Tetenov \(2018\)](#) studies empirical welfare maximization in richer policy classes. Here I minimize a confidence-region bound after observing the data. I do not establish a rule that minimizes worst-case expected regret over repeated samples. The contribution is the empirical accounting of diversification, selection, and inferential scope in an important cash-benchmarking experiment. The underlying treatment-choice logic is established; the empirical question is how much protection it supplies at the trial’s budget and inferential support.

The measured objective remains limited. Household dietary diversity describes current food access, while child nutrition, future resources, and public services may rank aid differently. [Currie and Gahvari \(2008\)](#) reviews reasons for in-kind provision; [Banerjee and Duflo \(2007\)](#) documents

spending priorities beyond food; and [Haushofer and Shapiro \(2016\)](#) studies cash effects on consumption, assets, and food security. The local price and nonrecipient responses in [Cunha et al. \(2019\)](#); [Egger et al. \(2022\)](#) further limit transport of an original-delivery lottery. An illustrative saving model in the appendix explains how intertemporal utility can rise without a current dietary gain. The bundled experiment supplies no variation identifying that mechanism separately.

2 Decision problem and theoretical framework

Let $a \in \mathcal{A} = \{0, g, \ell, m, u, L\}$ denote control, Gikuriro, and the four cash packages. Let $D_i(a)$ be the diet of baseline household i when its village receives assignment a , at the original within-village saturation. A higher value of a specified transformation $v(D)$ is preferred. I consider the mean $v(d) = d$, all integer survival indicators, and negative linear shortfalls

$$v_z(d) = -\frac{(z - d)_+}{z}, \quad z = 1, \dots, 12. \quad (1)$$

The thresholds are alternative priorities over a household food-access proxy. They are not official hunger cutoffs. The shortfall structure follows [Foster et al. \(1984\)](#); it does not assign nutritional equivalence to all food groups.

For fixed positive baseline weights w_i , define $\mu_a(v) = \sum_i w_i v(D_i(a)) / \sum_i w_i$ over the released baseline eligible sample. This target conditions on sample inclusion and its weights. Treating it as the mean of the entire eligible frame additionally requires the household sampling and observation assumptions. For policy deployment, I consider comparable villages with the same eligible-household weighting and package delivery. That transport restriction is maintained, not tested by random assignment.

A village lottery chooses $q_a \geq 0$ with $\sum_a q_a = 1$, independently of potential outcomes conditional on target blocks. With fixed provider costs c_a and $c_0 = 0$, the feasible set is

$$\mathcal{Q}(b) = \{q \geq 0 : \mathbf{1}'q = 1, c'q \leq b\}. \quad (2)$$

The source costs are standardized average costs per eligible household. In particular, cash overhead is costed separately as if each package served 56,127 households across eight districts, a national scale comparable to Gikuriro. These accounts do not observe the marginal cost of a small package share. Equation (2) conditions on their proportional allocation and stable delivery. It imposes an expected cost per eligible household; it does not ensure an exact realized cap for finitely many villages of unequal size.

Proposition 1 (Observed-package lotteries). *Under stable package outcomes and no cross-village interference, a lottery has $F_q(d) = \sum_a q_a F_a(d)$ and value $q'\mu(v)$. For known $\mu(v)$, a maximizer over $\mathcal{Q}(b)$ exists with at most two packages in its support.*

When $c_a < b < c_h$, the binding two-package vertex has

$$q_h = \frac{b - c_a}{c_h - c_a}, \quad q_a = 1 - q_h. \quad (3)$$

Every pure feasible package is also considered, including policies that leave funds unspent. The specified objective assigns no value to unused resources or benefits beyond the diet transformation. It therefore describes a hypothetical measured-diet benchmark, not an estimated institutional welfare function. Below I examine a binding expected-spending rule and alternative explicit values of unspent resources. At $b = c_g$, the full six-package set has nine vertices: pure Gikuriro and eight cash/control vertices. The latter are control, the three small pure cash packages, and four cheaper-package/large-cash mixtures. A budget change requires recomputing the vertices, including cost-order reversals.

Corollary 1 (Coverage comparison). *Universal Gikuriro exceeds the control/large lottery for objective v precisely when*

$$\mu_g - \mu_0 \geq (c_g/c_L)(\mu_L - \mu_0).$$

For mean diet, the difference equals $c_g\{\tau_g/c_g - \tau_L/c_L\}$, the rescaled parent-study cost-effectiveness comparison.

For uncertain arm means in a nonempty compact region $\mathcal{C}$, define

$$R(q, \mu) = \max_{r \in \mathcal{Q}(b)} (r - q)' \mu, \quad U_{\mathcal{C}}(q) = \sup_{\mu \in \mathcal{C}} R(q, \mu). \quad (4)$$

Regret measures lost expected value in the specified diet objective. If $\mathcal{C}$ covers the common true means, $U_{\mathcal{C}}(q)$ bounds their regret. A selected allocation may therefore use the same event without selecting a separate test after fitting it.

Proposition 2 (Diversification over a common region). *Let $\mathcal{V}$ be the vertices of $\mathcal{Q}(b)$. Then*

$$U_{\mathcal{C}}(q) = \max_{r \in \mathcal{V}} \sup_{\mu \in \mathcal{C}} (r - q)' \mu$$

is convex in q . A minimizer over $\mathcal{Q}(b)$ exists and need not be a vertex. If $\mathcal{C} = \{\mu : A\mu \leq d\}$ is compact, the minimum is obtained from a linear program with a nonnegative dual vector for each comparator vertex. On a simultaneous coverage event, its selected allocation has regret at most its reported bound.

For example, two treatments with costs below budget and independent mean intervals $[0, 1]$ have worst-case loss $\max(q, 1 - q)$ when q is the share given the first. The best vertex has loss one and the equal mixture has loss one half. Proposition 1 therefore limits the comparator under known means; it does not restrict the choice minimizing (4). This elementary illustration is established decision logic, not a new theorem about treatment allocation.

The fitted maximizer uses $\hat{\mu}$ as if known. The region minimizer instead protects against every mean vector admitted by $\mathcal{C}$. Neither procedure here is proved statistical minimax regret in the repeated-sampling sense. The region is an outer set rather than the sharp set of all compatible potential outcomes and distributions. Its bound can be conservative even when its coverage is valid.

3 Setting, sources, and measurement

The baseline survey took place in August–September 2016 and endline in September–October 2017. Gikuriro included village nutrition schools, farmer field learning schools, agricultural inputs, small livestock, Savings and Internal Lending Communities, and environmental health promotion. Eligibility combined poverty classification with a young child or a pregnant or lactating woman. Cash assignment operated at village level and covered the source’s eligible beneficiaries. Assignment effects include receipt, nonreceipt, and within-village consequences under original delivery.

The parent paper describes computer randomization in 22 district/poverty blocks of 10–13 villages. The released counts give 74 Gikuriro, 74 control, 22 each lower, middle, and upper cash, and 34 large-cash villages. Each small cash arm has one village in every block. The inspected public replication scripts contain no original assignment program establishing every randomization restriction. I assume labels are exchangeable conditional on the released block counts and assignments are independent across those blocks. Uniform quota assignment implies this model; additional assignment restrictions need not. The probabilities used below are $p_{ba} = n_{ba}/n_b$ under that explicit conditional model, rather than known solely because realized frequencies can be calculated. The public registry and its count discrepancy are documented in the replication design note; they do not resolve this missing assignment information.

I use the corrected CC BY 4.0 release of McIntosh (2025), DOI [10.5281/zenodo.15881329](https://doi.org/10.5281/zenodo.15881329). The earlier deposit was restricted after identifying fields were found. Its replacement explicitly reports their removal. The archive matches the provider’s checksum. The public numeric extract uses arbitrary replacement household, village, block, and child keys; it excludes contacts, geographic labels, dates, and beneficiary lists. These are processed research data. I do not reconstruct all upstream survey cleaning.

The household extract contains 5,506 rows. There are 1793 baseline eligible households, compared with 1,794 in the original paper, and 1751 eligible endline panel households. Complete endline food modules identify 1728 integer diets. Item-level information identifies two further diets, giving 1730. Table 1 reports the released assignment and initial complete-module sample; Table 13 accounts for the full baseline cohort.

HDDS counts twelve groups consumed by anyone in the household during the preceding day. The questionnaire’s sixteen categories are combined into twelve by taking any positive vegetable, fruit, or meat component for the respective group. Other groups remain distinct, including oils, sweets, and condiments. This follows the household score described by FAO (2011). It measures food access and variety, with a ceiling of twelve; it cannot establish calorie sufficiency, individual

Table 1: Village assignment, household observation, and provider costs

Arm	Cost (USD)	Villages	Baseline eligible	Observed diet	Retained (%)
Control	0	74	521	497	96.0
Gikuriro	124.49	74	540	522	97.4
Lower	53.58	22	165	160	96.4
Middle	95.86	22	154	148	97.3
Upper	121.24	22	167	164	98.9
Large	517.44	34	246	237	96.9

Notes: Costs are source USD per eligible household under the standardized accounting convention. Baseline weights expand the released sample; complete-module endline observation is distinct from item-informed diet identification. No treatment-receipt conditioning is used.

nutrient adequacy, or hunger severity.

The supplied endline score includes seven noninteger values near 4.463. Underlying responses can be missing or contain 888/999 codes. Complete binary modules match the reconstructed integer score exactly. For an incomplete module, a combined group is certainly positive if any component is positive and certainly zero only if every component is observed zero. The remaining uncertain groups define an interval $[D_i^-, D_i^+]$. Absent endline diets receive $[0, 12]$. The reconstruction retains this information instead of placing an imputed fractional score into an integer distribution. Source scores remain in the crosswalk and secondary sensitivity.

The supplemental household file provides source consumption, productive assets, saving, borrowing, health knowledge, sanitation practices, and purchased/own-produced food values. Economic source indices and inverse hyperbolic sine (IHS) transformations are inherited where provided. Both food-value fields use source weekly recall and a newly computed asinh transformation; the inspected labels do not recover a defensible monetary price base for conversion. I report transformation units rather than invented dollars or percentage changes. The individual panel provides standardized height-for-age, weight-for-age, and upper-arm scores, ages, and anthropometry flags. Those flags permit the child cohort accounting below. Every auxiliary outcome remains exploratory.

4 Empirical strategy and inferential scope

Table 2 separates the targets and their inferential events. The main allocation calculation concerns the released weighted baseline sample; observed-diet ratios and regression crosswalks remain sensitivities.

4.1 One distribution for each arm

For an identified transformation $Y_i = v(D_i)$ and its observation indicator R_i , define

$$\hat{\mu}_a = \frac{\sum_i T_{j(i),a} w_i R_i Y_i / p_{b(i),a}}{\sum_i T_{j(i),a} w_i R_i / p_{b(i),a}}. \quad (5)$$

Table 2: Targets, observation and inference

Target	Observation and weights	Inference and limits
Weighted released baseline	All baseline eligible households; fixed source weights and item intervals	Separate conditional 276-primitive and 276-term quota events; distribution and consistency restrictions. No wider-frame or deployment guarantee.
Potential identified diets	Package-specific potential reporting composition; positive weights	Separate 138-primitive event. Outcome observation can select different households by package.
Observed regression cross-walk	Selected endpoint outcomes; baseline controls and source weights	Village CR1/CR2 working approximations. No selection-robust baseline effect.
Block sensitivity	Same ratio/endpoint targets, 22 common block scores	Nominal coverage fails the stated synthetic diagnostics. Smaller bounds are unvalidated.
Deployment benchmark	Original packages and within-village saturation	Stable costs, delivery and cross-village spillovers are additional maintained assumptions.

Notes: Primitive counts refer to distinct fixed transformations and endpoints. Separate families do not imply one paper-wide coverage event.

The weights are positive and fixed at baseline. They numerically equal the eligible village frame count divided by its released baseline sample count, within floating-point precision. I invent no intensive-tracking factor. For identified diets, (5) targets the weighted potential-observation ratio under each arm. Different assignment-specific observation can change its composition. An observed-diet policy calculation is consequently a sensitivity measure; it represents policy diet for the baseline target only with an additional selection assumption.

For the baseline outcome, I replace Y_i by $v(D_i^-)$ or $v(D_i^+)$ and set $R_i = 1$ for every baseline eligible. Since every reported v is increasing, these endpoint means bound the common true arm mean. A contrast uses lower endpoints for positive coefficients and upper endpoints for negative coefficients to form its lower bound, with the assignments reversed for its upper bound. The resulting estimated endpoint region is subject to sampling uncertainty. It is not already a confidence interval for missing diets.

Applying (5) to indicators gives a normalized empirical CDF. Its means and shortfalls satisfy

$$\widehat{\mathbb{E}}[D] = \sum_{k=1}^{12} \widehat{\Pr}(D \geq k), \quad \widehat{\mathbb{E}}[(z - D)_+] = \sum_{k=0}^{z-1} \widehat{\Pr}(D \leq k)$$

exactly. The mean, twelve survivals, and twelve linear shortfalls have two affine duplicates, leaving 23 distinct transforms. The analysis covers increasing and increasing-concave priorities on the bounded integer score. Ratio normalization ensures coherence; it does not make finite-sample ratios exactly unbiased.

4.2 Finite conditional-assignment outer regions

Let $W_j = \sum_{i \in j} w_i$ be the fixed eligible baseline weight in village j , $W_b^{\max} = \max_{j \in b} W_j$, and $k_{ba} = n_{ba}$. For a primitive transformation supported on $[L, H]$, let $r = H - L$ and

$$V_a = \sum_b k_{ba} \left(\frac{W_b^{\max}}{p_{ba}} \right)^2, \quad t_a = r \sqrt{\frac{1}{2} V_a \log(2M/\alpha)}. \quad (6)$$

Here M is the number of primitive arm/endpoints protected together. If $\hat{D}_a$ is the denominator of (5), the interval $[\hat{\mu}_a - t_a/\hat{D}_a, \hat{\mu}_a + t_a/\hat{D}_a] \cap [L, H]$ covers its fixed conditional target with probability at least $1 - \alpha/M$. A zero denominator gives full support. Appendix A proves this through bounded sampling without replacement using Hoeffding (1963).

I use $M = 276$ for six arms, 23 transforms, and two full-baseline endpoints, with $\alpha = 0.05$. A separate identified-outcome family uses $M = 138$. These are separate 95 percent events. Their conjunction is not asserted to have 95 percent coverage. The same primitive event covers every transformation used in the coherent distributional projection below. Retaining only mean intervals would discard protected information. For an unobserved diet, the lower endpoint's lower confidence bound and upper endpoint's upper bound enclose the true mean. This finite statement conditions on the exchangeable assignment model, the fixed weights, and stable potential outcomes. It does not prove full-frame sampling representativeness, transport, or a hard budget guarantee.

4.3 Outcome consistency and coherent distributional regions

Write p_{ad} for arm a 's weighted potential probability at integer score $d = 0, \dots, 12$. For the baseline target, let L_{av} be the finite lower bound for $v(D^-)$ and U_{av} the finite upper bound for $v(D^+)$. Define

$$\mathcal{P}_a = \left\{ p_a \geq 0 : \sum_d p_{ad} = 1, \quad L_{av} \leq \sum_d v(d) p_{ad} \leq U_{av} \text{ for every protected } v \right\}. \quad (7)$$

Every true arm distribution satisfies these inequalities on the original 276-primitive event. The separate 138-target family can be used for potential identified-diet ratios because all their transformations use the same diet-observation indicator. A transformation known despite missing items does not convert an unidentified diet into an observed one.

Consistency gives additional deterministic restrictions. Let $W = \sum_i w_i$ and let S_a contain households whose village actually receives a . For increasing v with support $[L_v, H_v]$,

$$\frac{\sum_{i \in S_a} w_i v(D_i^-) + L_v \sum_{i \notin S_a} w_i}{W} \leq \mu_a(v) \leq \frac{\sum_{i \in S_a} w_i v(D_i^+) + H_v \sum_{i \notin S_a} w_i}{W}. \quad (8)$$

These inequalities use no assignment probabilities. They leave counterfactual outcomes unrestricted on their support and rely on consistency of the reported score with the observed food items. I impose them for every transformation in (7). A scalar consistency bound can be sharp while the resulting

joint moment polytope is only an outer set. I make no general joint-sharpness claim.

Proposition 3 (Using the protected distribution). *Suppose all protected primitive limits cover their arm targets simultaneously. Intersecting (7) with valid consistency restrictions preserves that event. For a specified v , minimize and maximize $\sum_d v(d)p_{ad}$ within each arm polytope. Their product encloses the common arm means without a new multiplicity penalty. It is the exact scalar projection of this arm-separable outer polytope. The selected allocation’s loss bound from Proposition 2 remains valid on the same event.*

The optimization fails explicitly if a distribution polytope is empty. It never drops conflicting constraints to produce a certificate. Each projection retains feasible endpoint distributions, and an independent epigraph program over the projected box’s 64 corners verifies the allocation calculation. This projection is a use of existing information, not a new confidence procedure. Moment-wise consistency and the finite event remain conservative, and other justified procedures may extract more.

4.4 A shared quota-moment region

The preceding construction first assigns a separate interval to every protected target and then projects their intersection. I also use a single exponential budget for all packages and transformations. For a fixed increasing score function $z(d) \in [0, 1]$, let y_j be its weighted potential village mean and u_a its full weighted baseline mean. For a uniform subset A_b of k_{ba} out of n_b villages, define the centered block residual

$$G_b(y, u_a) = \sum_{j \in b} \left(\frac{\mathbf{1}\{j \in A_b\}}{p_{ba}} - 1 \right) W_j(y_j - u_a).$$

The across-block sum equals the arm’s Horvitz–Thompson residual because $\sum_j W_j(y_j - u_a) = 0$. For any $\lambda_a \geq 0$, an outcome-free log-moment upper bound is

$$B_{ba}(\lambda_a) = \log \max_{y \in \{0,1\}^{n_b}, u \in \{0,1\}} \left(\frac{n_b}{k_{ba}} \right)^{-1} \sum_{|A_b|=k_{ba}} \exp\{\lambda_a G_b(y, u)\}, \quad B_a = \sum_b B_{ba}. \quad (9)$$

The maximum over binary vertices bounds every bounded village population by convexity. It deliberately relaxes the relation between the village outcomes and their true overall mean. Symmetry reduces the populations to sorted thresholds, and a positive dynamic program averages the quota subsets implicitly (Appendix D.2). This computes the relaxed moment without enumerating exponentially many subsets or claiming a sharp least-favorable potential-outcome population. Complementing y and u reverses the residual, so the same constant bounds its negative sign. This does not assert symmetry of any individual unequal-weight population’s sampling distribution.

For each of 23 distinct normalized score functions, let N_{az}^- and N_{az}^+ be the assignment-weighted

observed endpoint sums, and $\hat{D}_a$ the full-baseline assignment-weighted denominator. Define

$$e_{az}^+(p_a) = \exp\{\lambda_a(N_{az}^- - \hat{D}_a z' p_a) - B_a\}, \quad e_{az}^-(p_a) = \exp\{-\lambda_a(N_{az}^+ - \hat{D}_a z' p_a) - B_a\}. \quad (10)$$

Each term has expectation at most one at the true arm distribution, even when missing diets differ by assignment. Summing the 276 one-sided terms and applying Markov's inequality gives the following event.

Proposition 4 (Conditional quota mixture). *Maintain uniform quota assignment within blocks, independent blocks, fixed positive baseline weights and consistent bounded potential diets. With probability at least 0.95, the true six score distributions lie in*

$$\mathcal{C}_Q = \left\{ (p_a)_a : p_a \in \mathcal{P}_a^{\text{obs}}, \quad \sum_{a,z,\sigma \in \{+,-\}} e_{az}^\sigma(p_a) \leq 5520 \right\},$$

where $\mathcal{P}_a^{\text{obs}}$ imposes unit mass, nonnegativity, deterministic outcome consistency and observed identified-score bin floors. Every data-selected feasible allocation has regret bounded by a valid support upper bound over this common region.

The exponential scale is chosen on the fixed grid $\{1, 1.25, 1.5, 2\} \sqrt{8 \log(5520)/V_a}$ by minimizing $\{B_a(\lambda) + \log(5520)\}/\lambda$. Only baseline weights, quotas and the stated family enter that choice. Allocation shares and support dual variables may use observed outcomes because Proposition 4 protects every allocation on one event. The new event is separate from the earlier 276-primitive and 138-identified-target events. I do not intersect their nominal 95 percent sets and retain that level.

The code interprets normalized score entries as fixed stored bounded functions, encloses moment and observation arithmetic outward, and projects the resulting probability region onto the exact mean and six-group shortfall objectives. For each of the nine ideal cost vertices, a nonnegative Lagrange multiplier splits the shared exponential budget into six convex arm problems. Enclosed tangents and feasible LP duals give support upper bounds regardless of whether the numerical candidate minimizes its convex problem. The reported shares are outcome-selected exchange-search proposals with verified loss bounds; their search gaps are numerical diagnostics, rather than certified minimax optimality. Appendix A gives the coverage and numerical-bound arguments. The package retains the proposal, constants, tangents and dual witnesses for direct replay.

4.5 Exploratory block approximation and stress tests

For block b and arm a , let N_{ba} and D_{ba} be the numerator and denominator contributions in (5). The ratio score is

$$u_{ba} = \frac{N_{ba} - \hat{\mu}_a D_{ba}}{\sum_h D_{ha}}, \quad \hat{\Sigma} = \frac{B}{B-1} \sum_b u_b u'_b, \quad B = 22. \quad (11)$$

Table 3: Finite-block stress diagnostics for the exploratory approximation

Scenario	Covered	Replications	Joint coverage	MC lo.	MC hi.
balanced equal weights	695	800	0.869	0.843	0.891
block heterogeneity	695	800	0.869	0.843	0.891
unequal frame weights	592	800	0.740	0.708	0.770
rare positive effect	800	800	1.000	0.995	1.000

Notes: Synthetic fixed potential outcomes, independent exchangeable assignments, and the 22 released block quotas. Coverage protects fifteen arm-pair mean contrasts, rather than the actual 25-transform endpoint family. MC intervals are exact binomial intervals for simulated coverage. They are neither actual Rwanda coverage estimates nor evidence that nominal 95 percent coverage has been attained.

Grouping before covariance retains dependence among all arms assigned within a block. Arm-exclusive village scores would force cross-arm covariance to zero. Appendix A gives an exact counterexample: that omission estimates only 90 percent of the true difference variance in a ten-village quota design, even as the number of identical blocks increases. The common block covariance repairs this omission. Its asymptotic justification additionally requires increasing independent blocks, stable positive denominator limits, and no dominating block.

Shared block Rademacher multipliers generate exploratory maximum-studentized bands with 99,999 draws. A 99 percent Monte Carlo upper order statistic protects the estimated 95 percent multiplier quantile from simulation error within each family. That Monte Carlo protection does not address sampling undercoverage. I retain separate 184 Gikuriro/cash, 828 all-policy, and 1656 endpoint families, deduplicating affine repetitions. A zero estimated variance receives its deterministic support range rather than a structural-zero interval; 121 displayed rows require this guard. Pointwise intervals use 21 block degrees of freedom. None of these choices establishes exact weak-null inference.

Synthetic tests use the actual released quota structure, 800 repeated assignments per scenario, and a 4,999-draw diagnostic multiplier calculation for fifteen mean comparisons. Table 3 reports joint coverage. The equal-weight and unequal-weight failures remain material even with Monte Carlo upper quantiles. The rare-effect scenario obtains full-support intervals when its realized variance vanishes. Its full coverage cannot rescue the other scenarios. I therefore label every resulting block-region allocation and loss bound as exploratory. ANCOVA CR2 diagnostics concern a different estimator and cannot validate these ratios or missingness endpoints.

5 Results: diet comparisons and feasible allocations

5.1 Assignment contrasts allowing incomplete diets

Large cash has a higher observed mean than control. The initial ANCOVA estimate is 0.545 groups, and the reconstructed parent specification reproduces the original paper’s published rounding (Appendix Table 14). That is an assignment comparison among measured outcomes. In the

Table 4: Mean dietary comparisons for the weighted baseline sample

Comparison	ID lo.	ID hi.	Approx. lo.	Approx. hi.	Finite lo.	Finite hi.
Large vs. control	0.171	0.731	–	–	-11.257	12.000
GK vs. Control + Large	-0.275	0.323	-0.703	0.686	-11.436	11.567
GK vs. Lower + Large	-0.574	0.101	-1.113	0.684	-12.000	11.567
GK vs. Upper + Large	-0.293	0.005	-0.771	0.437	-12.000	11.567

Notes: Units are food groups. Positive values favor the first policy. ID columns are estimated missing-diet endpoint regions, not confidence limits. Approximate columns are the exploratory block endpoint bands, which failed stress diagnostics. Finite columns derive from the simultaneous 276-primitive conditional-assignment family. Large cash is unaffordable as a pure equal-budget policy, so its direct comparison is not included in the nine-policy multiplier family; no such approximate band is invented.

full weighted baseline sample, the item-informed estimated region for large cash minus control is $[0.171, 0.731]$ groups. Its positive sample endpoints are consistent with a full-baseline improvement, but the finite conditional envelope $[-11.257, 12.000]$ is wide and includes zero. I consequently do not claim an established positive full-baseline effect from these bounds.

For the feasible budget comparisons in Table 4, Gikuriro’s estimated mean region crosses zero against each reported cash lottery. Against lower/large cash, the observed-diet ratio difference is -0.237 groups. The sign remains unresolved after allowing incomplete diets and inferential uncertainty. A lottery over observed packages makes the untreated fraction explicit; it does not identify the outcome of giving every household a transfer equal to the lottery’s average cash amount.

One common diet distribution also permits comparisons across every survival threshold and linear shortfall. The estimated six-group shortfall endpoint comparisons can favor cash, while their uncertainty envelopes include zero. Neither increasing nor increasing-concave dominance is established by the finite region. Oils, sweets, and condiments remain components of the score, so even a uniform distributional ranking would concern measured household food access rather than individual nutrition.

5.2 Diversification and information from the outcomes

Lower/large cash has the highest fitted mean across the nine original-budget vertices. Its identified-diet mean exceeds Gikuriro by 0.237 groups. Agreement across regression adjustments does not identify the best allocation for all baseline households. Table 5 instead asks how much the observed information tightens a common worst-case loss calculation.

The support-only mean bound is 9.600 . Equal probability on control, Gikuriro and the three small cash packages attains it at expected expenditure $\$79.03$. At least one of those five pure feasible packages has share no greater than one fifth under any allocation. Giving it mean twelve and every other package mean zero produces loss at least $12(1 - 1/5) = 9.6$. Thus much of the reduction from a single package’s twelve-group worst case follows from diversification alone.

The observed consistency restrictions lower the bound to 8.074 . A finite mean box alone gives 9.440 , whereas using its protected survival and shortfall information gives 8.151 . Intersecting

Table 5: Diversification and incremental outcome information

Information region	Mean loss (groups)	Shortfall loss
Support and costs only	9.600	0.800
Outcome consistency	8.074	0.666
Finite primitive box	9.440	0.715
Finite, coherent distribution	8.151	0.687
Finite and consistency	7.720	0.637
Quota mixture, verified proposal	5.868	0.503

Notes: Same original-budget comparator menu and full weighted baseline target. Support-only uses no outcomes; consistency uses assigned item intervals. The first three finite rows inherit one 276-primitive event. The quota row uses a separate 276-term conditional event and verifies a numerical allocation proposal, rounding its upper bounds upward. Its exact minimax optimum is not claimed. Mean loss is in groups; shortfall loss is on a unit-range objective. No joint 95 percent coverage across the separate procedures is asserted.

Table 6: Diversified allocations by information region

Region	Control	Gikuriro	Lower	Middle	Upper	Large	USD
Support and costs only	20.0	20.0	20.0	20.0	20.0	0.0	79.03
Outcome consistency	16.5	14.3	23.3	22.5	23.3	0.0	80.20
Finite, coherent distribution	14.9	11.0	25.1	23.7	25.3	0.0	80.59
Finite and consistency	14.4	10.3	25.5	24.0	25.7	0.0	80.75
Quota mixture proposal	10.4	5.8	29.3	24.9	29.2	0.4	84.27

Notes: Percent village shares; expected USD per eligible household. Support-only is a no-outcome benchmark. Other rows use the weighted baseline target. The first four rows minimize their specified box bounds; the quota row is a verified numerical proposal. The objective leaves unspent resources unvalued. None is an operational recommendation or an estimate of household preferences.

coherent finite and consistency restrictions yields 7.720, a total tightening of 1.880 groups relative to support alone. Along the nested sequence from support to consistency to their finite intersection, consistency supplies 1.526 groups of tightening and finite inference adds 0.353 groups, or 4.38 percent of the preceding bound. This accounting depends on the sequence: the finite-only and consistency regions are not nested. It is a diagnostic of that procedure, rather than a general valuation of conducting an experiment.

The quota mixture gives a verified mean-loss bound of 5.868 and a six-group shortfall bound of 0.503. The mean bound is 24.0 percent below the earlier coherent finite construction. Using the same data with a stronger justified event therefore changes the decision’s numerical protection materially. It still allows a loss of nearly six food groups and about half the shortfall objective’s range. The bounds cannot support a small-loss recommendation. The separate events need not be nested, and the displayed difference is not an estimated effect of collecting data. Figure 1 shows both constructions.

The intersected mean allocation spends \$80.75 and retains a control share. Its diffuse probabilities reflect protection against broad counterfactual uncertainty, not estimated positive effects of every package. The exploratory block region permits a much smaller bound, 0.570, but its stress-test failures remain. I do not interpret that figure as a reliable 95 percent guarantee or calibrate

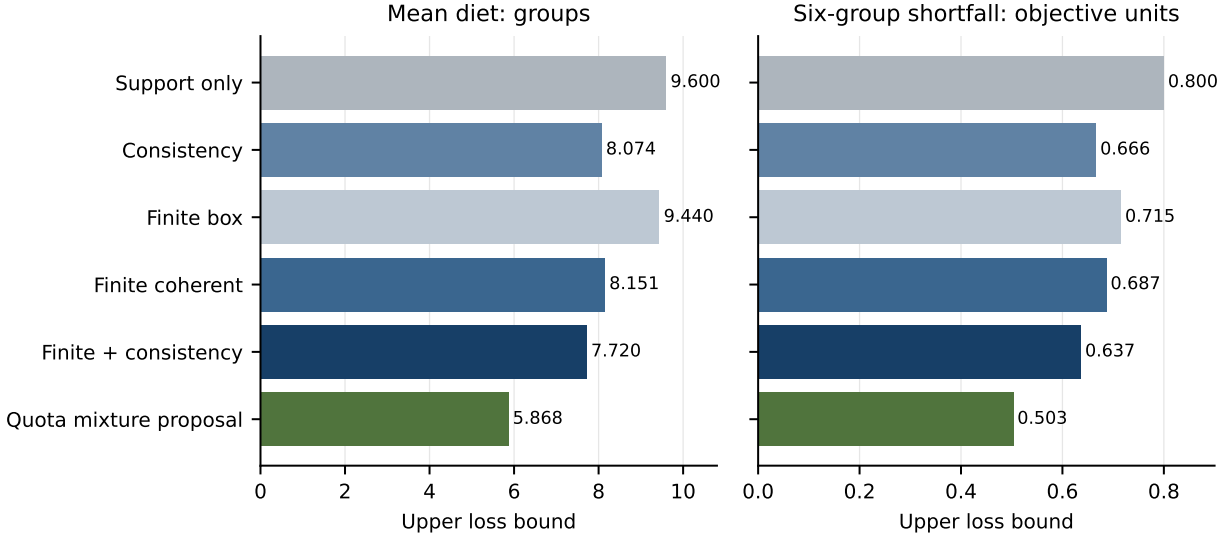


Figure 1: How the information region changes the loss bound

Notes: Same objective and feasible menu within each panel. The finite mean box and consistency region are not nested. The quota event is a separate construction retaining the shared exponential budget; its bar verifies a selected allocation rather than certifying exact minimax optimality. Bar differences do not estimate the general value of experimental information.

an inflation factor until selected simulations pass.

The quota proposal spends \$84.27. It assigns approximately 10.4 percent to control, 5.8 percent to Gikuriro, 29.3 percent to lower cash, 24.9 percent to middle cash, 29.2 percent to upper cash and 0.4 percent to large cash. These shares differ sharply from the fitted lower/large winner because the protected distributions still allow several affordable packages to outperform it. The master checks exact rational shares and costs, then verifies the loss bounds through the nine comparator vertices. It does not infer a positive effect for each chosen package from its share.

5.3 What changes the guarantee?

I compare the quota construction with established exponential bounds under the same weighted target, item-endpoint rules, score functions, costs and rational allocation proposals. Appendix D includes without-replacement Hoeffding, Serfling, an unrelaxed harmonic martingale bound, a weight-sensitive product bound, and an empirical-Bernstein adaptation with a fixed baseline forecast. Each alternative selects its statistical scale from baseline quantities and uses its own shared event. I separately choose support multipliers and verify the upper certificates. The inexpensive product/harmonic hybrid gives 6.609 groups and 0.557 shortfall units, compared with 5.868 and 0.503 for the quota procedure. Independent exact-rational adversarial distributions and a feasible policy dual now bracket the optimized loss within each implemented outer region. For quota, the mean bracket is $[5.866228, 5.867129]$ groups: the selected proposal is within 0.000901 groups of the regional minimax value. Thus its broad guarantee is not explained by a poor allocation search within that relaxation. This is a result about the implemented region, whose distributions need

Table 7: Regret protection under hypothetical institutional rules

Hypothetical rule	Own-menu loss	Common-menu loss	USD, common-menu optimum
Expected-cost ceiling	7.720	7.720	80.75
At least 75 percent assisted	7.522	7.720	80.75
Universal assistance	7.503	8.138	121.94
Binding expected spending	7.392	7.848	124.49
Universal, binding spending	7.291	8.145	124.49
At least 25 percent GK	5.790	7.940	90.15

Notes: Coherent finite-and-consistency mean region. “Own-menu” minimizes loss when both allocation and comparator obey the stated rule. “Common-menu” constrains the choice but retains the original unrestricted comparator; its separately optimized expenditure is shown. The columns therefore need not select the same allocation. Constraints and costs are hypothetical benchmarks, not measured donor preferences or mixed-rollout costs.

not correspond to a jointly sharp finite potential population. It proves neither a lower bound on the best possible inference procedure nor a small-loss deployment conclusion.

An ablation using the same quota moments but separate tail inequalities gives 6.878 groups and 0.577 shortfall units. Thus, retaining the shared exponential constraint matters beyond improving the moment constants. Removing identified-bin floors barely changes the displayed bounds while retaining scalar consistency. A deterministic design catalogue also shows that moment gains depend on weight concentration: in twelve-village half-quota blocks, the scalar radius decreases by about 2.3 percent with equal weights and 49.2 percent when one village has weight ten and the others weight one. This limited comparison neither establishes dominance over variance-adaptive inference nor measures the value of field information.

5.4 Assistance requirements and the comparator

CRS implemented the bundled programme; the experiment does not reveal a CRS mandate to optimize only HDDS or to randomize the listed packages at scale. I therefore treat (2) as a benchmark rather than an observed organizational rule. Table 7 makes several alternative hypothetical rules explicit. A minimum assisted share excludes some control allocations; universal assistance excludes them all. Here assistance means village assignment or an offer under the original participation pattern, rather than verified receipt by every household. A binding expected budget is different from a hard realized spending cap. Every own-menu calculation changes both choice and comparator to the stated class.

Universal assistance increases the optimized worst-case loss bound against the common menu from 7.720 to 8.138. Binding expected expenditure gives 7.848. In contrast, own-menu bounds can fall because alternatives are excluded from the comparator. Those smaller numbers do not establish that a restriction improves measured outcomes. This comparison also differs from the actual dietary opportunity cost at a fixed true mean vector. For a restricted choice class $\mathcal{K} \subseteq \mathcal{Q}$, that cost is $\max_{r \in \mathcal{Q}} r' \mu - \max_{q \in \mathcal{K}} q' \mu$. The difference between separately optimized worst-case bounds is another object. All six means equal to five are admitted by the displayed outer region; at that vector every nonempty restriction has zero actual dietary opportunity cost, even though its

optimized regret bound can be larger.

Corollary 2 (A fixed package share and a changing comparator). *For $0 \leq \gamma \leq 1$ and $c_g = b$, imposing $q_g \geq \gamma$ on both allocation and comparator makes the feasible set $\gamma e_g + (1 - \gamma)\mathcal{Q}(b)$. For any fixed common mean region, the minimized own-menu regret is $(1 - \gamma)$ times the original minimum.*

For example, a 25 percent minimum Gikuriro share gives own-menu loss 5.790, exactly three quarters of 7.720. This is a geometric consequence of the comparator restriction, rather than empirical evidence favoring that mandate. Keeping the original comparator prevents this mechanical reduction from being interpreted as improved protection against foregone alternatives.

Unused expected funds require a separate choice of objective. The original criterion contains only current diet. A hypothetical value η groups per unspent dollar changes the value to $q'\mu + \eta(b - c'q)$ and changes comparator advantages accordingly. Appendix Table 10 reports several explicit values. They are not estimated welfare weights or evidence that an organization should withhold funds. Additional benefits, fiscal opportunity costs or obligations can change the allocation even if the measured diets remain fixed.

5.5 Cost accounting and magnitude

The simplest back-of-the-envelope calculation is reach: the \$124.49 budget finances only 24.1 percent large-cash coverage at its \$517.44 standardized cost. Combining lower with large cash instead uses 15.3 percent large packages and reaches every village. These shares follow from source costs and the lottery definition. They do not assume a linear household response to dollars. Changes in prices, implementation scale, or participation may alter both costs and outcomes.

I rebuild the policy set in 32 accounting scenarios: a five-by-five grid of Gikuriro and cash cost factors, three common-take-up settings, and four Gikuriro avertable-cost settings. The source decomposition is $c_a^{elig} = c_a^{benef}(1 - s_a + s_a r_a)$, where s_a is the avertable share and r_a compliance. The scenarios use coherent Hájek mean point estimates and the earlier coherent finite and finite-and-consistency regions. They no longer substitute initial ANCOVA means for the headline decision. Outcomes remain fixed as costs change, making these accounting sensitivities rather than treatment-delivery forecasts.

Upper cash costs only \$3.25 less than the original budget, so relative-cost changes can remove its pure vertex. This affects the fitted observed policy and its competitors even before uncertainty enters. Genuine fixed activation costs, $F_a \mathbf{1}\{n_a > 0\} + m_a n_a$, create a different feasible set. The nationally standardized average-unit accounts do not identify those costs for a small mixed deployment. I therefore condition on the stated allocated-cost convention rather than invent a hard-budget solution.

Table 8: Diversified mean-loss bounds under selected cost scenarios

Accounting scenario	Upper-cost margin (USD)	Finite coherent loss	With consistency
GK costs -25%	-27.88	7.606	7.205
GK costs $+25\%$	34.37	8.170	7.738
Cash costs -25%	33.56	8.176	7.744
Cash costs $+25\%$	-27.06	7.725	7.322

Notes: Every scenario reenumerates vertices and optimizes all six allocation shares. The upper-cost margin is budget minus pure upper-cash cost; a negative value removes that pure package from the feasible set. Loss is in food groups. The full file retains all 32 cases for both coherent finite regions. Outcomes remain fixed; these are not cost confidence intervals or forecasts for changed take-up.

6 Robustness and interpretation

6.1 Regression crosswalk and weight sensitivity

The appendix compares the original-style ANCOVA with the current specification. Complete-case families, coarse missing-outcome calculations and influence diagnostics remain in the replication tables; they are auxiliary checks rather than additional main findings. The regression conditions on baseline transformations, a missing indicator, and assignment blocks. It imputes missing baseline controls by the weighted within-block mean and never fills a missing endline outcome for the integer analysis. Village CR1 standard errors and maximum-studentized bands are working asymptotic assessments. CR2/Satterthwaite corrections use a specified inverse-weight working covariance. These estimators have different scopes from the block ratio and finite conditional calculation.

The source-pooled crosswalk reproduces the original study’s rounded Gikuriro and large-cash mean coefficients. Subsequent columns split the smaller cash arms, remove the original source-selected control list, and change the outcome and baseline measurement. No new LASSO selection is run. Equal-household and equal-village specifications also change which households or villages receive representation. Their signs provide composition sensitivity rather than independent estimates of the same weighted target. Weight concentration and village/block omission diagnostics expose influential units without making the remaining ratios selection robust.

6.2 Baseline child cohort and auxiliary outcomes

The endline anthropometry sample combines children with different cohort histories. Source flags identify 2265 baseline-eligible children flagged for baseline anthropometry. Of these, 2213 have a linked endline row. The source endline eligibility/due flags identify 3017 children, of whom 2211 belong to that baseline cohort and 806 do not. The latter are outside the baseline-flagged cohort; the flags do not show that every one is a newborn. Table 9 reports the partition by assignment.

To assess cohort composition, I retain those baseline children regardless of endline eligibility or due status and estimate follow-up score regressions using their baseline assignment, weights, and source lagged scores. Missing follow-up scores still select observations. No estimate in the separate fifteen-test Holm family is significant at five percent (Appendix Table 12). This does not identify

Table 9: Child cohort accounting under source anthropometry flags

Arm	Baseline cohort	Linked endline	Measured endline	All endline due	Outside cohort
Control	676	654	628	893	239
Gikuriro	672	659	650	913	254
Lower	199	193	189	262	69
Middle	196	190	187	253	63
Upper	204	204	199	287	84
Large	318	313	303	409	97

Notes: Baseline cohort membership is pre-treatment: eligible households and source baseline-anthropometry flag. Linked and physically measured endline counts differ from nonmissing standardized follow-up scores. “Outside cohort” concerns membership in this measured cohort, not identified births. Assignment and weights are inherited from baseline households.

a zero full-cohort nutritional effect or replace the parent’s prespecified child analysis. The original exploratory endline-due regressions and ineligible-household estimates remain in Appendix Table 15; neither establishes zero nutrition benefits or zero spillovers.

The auxiliary saving-stock regression gives a Gikuriro coefficient of 1.186 IHS units, with standard error 0.338 (Appendix Table 11). Its pointwise p -value is < 0.001 ; Holm adjustment across the economic outcomes gives 0.052. The response is compatible with a household shifting resources toward future consumption. Appendix B develops that possibility: a higher return to saving can raise intertemporal utility and reduce current food spending, while discrete group thresholds leave variety unchanged. Testing this explanation would require variation in saving-group access or terms separately from the nutrition, agriculture and sanitation activities. The bundled assignment supplies no such comparison, irrespective of the saving coefficient’s significance.

6.3 Deployment and omitted value

The deployment benchmark holds eligibility, participation and package delivery at their trial values. A different mix of villages could change prices, networks or provider capacity. Those responses enter the allocation problem through outcomes and costs; estimating them would require variation in coverage beyond the original assignments. A donor’s minimum-assistance rule would instead change the feasible menu directly, as in Table 7.

For benefits outside current diet, let H_a denote an additional package value in diet-equivalent units. Gikuriro exceeds lottery q under that broader criterion when $H_g - \sum_a q_a H_a \geq -(\mu_g - q'\mu)$. At the fitted identified-diet means, the required differential relative to lower/large cash is 0.237 groups. This gives an accounting threshold for a specified extra benefit. Health outcomes, saving returns and preferences would be needed to estimate it; selection uncertainty already permits the full-baseline diet comparison to reverse sign.

7 Discussion and conclusion

At Gikuriro’s expected provider budget, a lottery combining lower and large cash has the highest fitted mean among identified diets. It exceeds Gikuriro by 0.237 food groups. Assigning villages to that lottery maintains broad coverage and concentrates the larger package in 15.3 percent of them. The comparison uses the original participation and average-cost convention; it does not establish the dietary ranking for households whose follow-up diets remain unidentified.

The uncertainty calculation is much less decisive than the fitted ranking. Equal shares on the five individually affordable packages attain a 9.600-group bound using support and costs alone. Outcome consistency supplies 1.526 groups of tightening; the earlier conditional finite intersection adds 0.353. Using the quota moments and a shared exponential budget instead verifies a 5.868-group allocation bound. This reduces the earlier 7.720 bound substantially with the same released data, but still allows large losses. The shortfall bound of 0.503 is also broad relative to its unit range. These procedures establish neither a small-loss decision nor a lower limit on what every valid analysis could learn. The much smaller block approximation fails its reported coverage diagnostics.

The verified quota allocation assigns some villages to control and leaves expected resources unused. The institutional sensitivity on the earlier coherent region shows that universal assistance raises optimized regret against the original menu to 8.138. A donor must specify which packages and uses of funds are admissible before applying either calculation. Changing that menu also changes the comparator; the difference between optimized bounds cannot measure the actual dietary cost of a mandate at the unknown true means.

Gikuriro’s stated objective concerns maternal and child nutrition, whereas the main outcome records household food-group variety. The child follow-up regressions and saving illustration address parts of that gap but do not identify the full nutritional or intertemporal benefit of the package. Repeated dietary measures and separate variation in saving terms or program saturation could examine those responses. Original assignment documentation, provider objectives and rollout costs would also make the decision benchmark more informative. The current evidence supports conditional dietary comparisons and quantifies the remaining loss bound; it leaves the wider welfare ranking open.

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A Decision and inference proofs

Proof of Proposition 1. The distribution and objective follow from conditioning on package assignment under stable delivery and no cross-village interference. The feasible set is a compact polytope. A linear objective attains its maximum at a vertex. If the cost constraint is slack, the probability simplex permits only one positive coordinate at a vertex. If it binds, three positive coordinates allow a nonzero perturbation satisfying both the sum and cost equalities. A sufficiently small positive or negative perturbation preserves nonnegativity, contradicting extremality. Thus no vertex requires more than two packages. Equal costs or an exactly affordable pure package do not alter this upper bound. Corollary 1 follows by subtracting the control mean and using $q_L = c_g/c_L$.

Proof of Proposition 2. For each μ , maximizing $(r - q)'\mu$ over r is a linear problem whose maximum occurs at a vertex. A finite maximum commutes with the supremum over $\mathcal{C}$. Each resulting supremum of affine functions is convex in q . Compactness gives continuity and existence of a minimizer. For comparator vertex r_j , linear-program duality gives

$$\sup_{A\mu \leq d} (r_j - q)'\mu = \min_{\lambda_j \geq 0: A'\lambda_j = r_j - q} d'\lambda_j.$$

The allocation program minimizes t subject to $q \in \mathcal{Q}(b)$ and these dual equalities with $d'\lambda_j \leq t$ for every comparator. Its constraints are linear jointly in q, t, λ_j . When the common true μ belongs to $\mathcal{C}$, the bound holds for every q at once, including the data-selected minimizer. This coverage implication does not optimize expected regret over sampling distributions. The code independently solves the primal adversary for every comparator to check dual values and allocation feasibility.

Proof of Proposition 3. On the protected event, each true primitive expectation lies in its stated interval. Each arm's true probability vector is nonnegative, sums to one and satisfies (7). Consistency bounds follow by bounding each unassigned household's potential transformed score by $[L_v, H_v]$ and each assigned score by its item interval. Their intersection therefore retains the true vector. Because the defined distribution polytope has no cross-arm constraints, its scalar projection is the Cartesian product of the armwise minimum/maximum intervals: compactness attains each endpoint, and convexity attains every intermediate mean. This is exact projection of the stated outer set, not a claim that each vector corresponds to a finite potential-outcome population with the given household weights. The original simultaneous event covers all transformations before projection, so no additional multiplicity adjustment is needed. Proposition 2 then bounds regret for the selected allocation on that same event.

For the potential identified-diet ratio, write N_a and D_a for the known weighted numerator and denominator among assigned households with identified diets, and U_a for total unassigned baseline weight. Bounded unknown potential scores and observation indicators imply bounds $(N_a + L_v U_a)/(D_a + U_a)$ and $(N_a + H_v U_a)/(D_a + U_a)$ when this denominator is positive. Both endpoints are attained in the unrestricted scalar relaxation by observing all unassigned households at a support

endpoint. Assigned unidentified diets remain excluded even when a particular transform is known. These scalar inequalities are additional valid outer restrictions, not joint-sharpness or common-population selection assumptions.

Proof of Proposition 4. For fixed W_j , $G_b(y, u)$ is affine jointly in y and u . Its average exponential over quota subsets is convex on $[0, 1]^{n_b+1}$. Successively maximizing each coordinate on its interval shows that a maximum occurs at a binary vertex. Equation (9) therefore bounds the positive moment of every centered block residual. Replacing (y, u) by $(1-y, 1-u)$ reverses G_b , so it also bounds the negative moment. Independent blocks multiply their moment bounds. The population mean identity cancels the across-block centering, leaving the arm’s realized Horvitz–Thompson residual S_{az} . Consequently $\mathbb{E} \exp\{\pm \lambda_a S_{az} - B_a\} \leq 1$.

For every assignment, monotone item intervals imply

$$N_{az}^- - \hat{D}_a u_{az} \leq S_{az} \leq N_{az}^+ - \hat{D}_a u_{az}, \quad u_{az} = z' p_a^{\text{true}}.$$

Thus both computable terms in (10) are no greater than their corresponding true-residual terms. Each has expectation at most one without an assumption that missing diets are ignorable. Dependence among arms, transformations or their missingness patterns does not affect the expectation of their sum. Markov’s inequality gives probability at most $276/5520 = 0.05$ that it exceeds 5520. The true score distributions also satisfy unit mass, nonnegativity, consistency and a bin floor equal to the fixed-weight share actually assigned to that arm with an identified score. Intersecting these deterministic restrictions preserves coverage. On that one event, the common true distributions bound regret for every feasible q and every specified linear score objective. Data-dependent selection of shares or support dual variables incurs no further test selection.

Outward constants and support verification. For numerical evaluation of (9), the sorted-threshold recurrence uses the actual stored weights. Directed Decimal arithmetic encloses the exponent arguments, correctly rounded elementary functions are enlarged by adjacent Decimal values, and positive recurrence coefficients propagate the upper bounds. Appendix D.2 gives the construction. The earlier exhaustive integer-proxy implementation remains available as a separate reference, including its explicit weight-drift allowance; the current calculation needs no proxy rounding. Directed assignment-weight arithmetic supplies a lower endpoint numerator, an upper endpoint numerator and denominator enclosures. The stored affine log tests are no greater than the conceptual tests for any nonnegative probability vector. The normalized score entries themselves are fixed bounded functions; no exact floating-point identity with the raw objective is presumed.

Write $F_a(p_a)$ for the stored arm sum and $C = 5520$. For any $\tau \geq 0$ and direction d ,

$$\sup_{\sum_a F_a \leq C, p_a \in \mathcal{P}_a^{\text{obs}}} d' p \leq \tau + \sum_a \sup_{p_a \in \mathcal{P}_a^{\text{obs}}} \{d'_a p_a - \tau F_a(p_a)/C\}.$$

The negative arm objective is convex. At any candidate, interval-enclosed values and gradients

produce an affine tangent lower bound, with its gradient enclosure error bounded over $0 \leq p_{ad} \leq 1$. The tangent's minimum is bounded below using a feasible LP dual, rather than a numerical primal objective minus an arbitrary cushion. If the logical polytope is $p \geq f$, $Ap \leq b$, $\mathbf{1}'p = 1$, any $\ell \leq 0$ and ζ satisfying $A'\ell + \zeta\mathbf{1} \leq g$ give

$$\min g'p \geq g'f + \ell'(b - Af) + \zeta(1 - \mathbf{1}'f).$$

The code clamps suggested inequality multipliers to nonpositive values, constructs an outward feasible equality multiplier and evaluates this expression downward. It encloses score-direction rounding and uses exact rational allocation shares and ideal cost vertices. Taking the largest of the nine resulting support upper bounds verifies the selected allocation. Upward decimal rounding gives the displayed values. None of these steps relies on global success of the numerical minimizer or proves that the selected allocation is the exact minimax solution. The audit replays all tangents and checks dual feasibility independently with exact fractions.

Proof of Corollary 2. For $\gamma < 1$, every q with $q_g \geq \gamma$ can be uniquely written as $\gamma e_g + (1 - \gamma)s$. Nonnegativity and the sum restriction hold exactly when s is a simplex vector. Since $c_g = b$, its cost inequality is equivalent to $c's \leq b$. The same transformation applies to each comparator, so their difference is $(1 - \gamma)(r - s)$. Taking the supremum and minimum scales the regret by $1 - \gamma$. For $\gamma = 1$, the class contains only Gikuriro and own-menu regret is zero. Nothing in this identity ranks restricted policies against excluded alternatives.

Finite conditional-assignment interval. Fix an arm, a primitive outcome supported on $[L, H]$, and its target $\mu = \sum_i w_i R_i(a) Y_i(a) / \sum_i w_i R_i(a)$, assuming a positive potential denominator. For complete baseline endpoints $R_i(a) = 1$. Define village residuals

$$X_j(\mu) = \sum_{i \in j} w_i R_i(a) \{Y_i(a) - \mu\}.$$

Because $\mu \in [L, H]$, every residual in block b lies in $[(L - \mu)W_b^{\max}, (H - \mu)W_b^{\max}]$, including villages with unobserved outcomes or a smaller total weight. This interval has width $(H - L)W_b^{\max}$. Its inverse-probability-scaled width is $(H - L)W_b^{\max}/p_{ba}$.

Under exchangeable quota assignment, the arm's residual sum in a block is a simple random sample of size k_{ba} from the fixed village residuals divided by p_{ba} . Theorem 4 of [Hoeffding \(1963\)](#) bounds the convex exponential moment of this without-replacement sum by the with-replacement moment. Applying the bounded-variable exponential inequality to those k_{ba} independent draws, and multiplying across independent blocks, gives

$$\Pr \left(\left| \sum_j \frac{T_{ja} X_j(\mu)}{p_{b(j),a}} \right| > t \right) \leq 2 \exp \left\{ -\frac{2t^2}{(H - L)^2 V_a} \right\}.$$

The total residual expectation is zero since $\sum_j X_j(\mu) = 0$. Substituting (6) makes this failure probability at most α/M . The realized residual is $\hat{D}_a(\hat{\mu}_a - \mu)$. Dividing by its positive realized denominator therefore yields the reported ratio interval. This inversion tests the fixed true μ ; it needs no union over a grid of candidate means. A zero realized denominator receives full support. A union bound protects the M fixed primitive arm/transformation/endpoints. Applying deterministic endpoint ordering then covers true missing-diet means and every policy computed from the same primitive region. Dependence across arms is allowed in the union bound; independent blocks and conditional exchangeability remain essential. The bound is classical and conservative, not an optimal confidence procedure.

Block covariance and the referee counterexample. For independent population linearization scores ψ_b with means m_b satisfying $\sum_b m_b = 0$, the expectation of their centered block covariance is

$$\mathbb{E} \left[\frac{B}{B-1} \sum_b (\psi_b - \bar{\psi})(\psi_b - \bar{\psi})' \right] = \sum_b \text{Var}(\psi_b) + \frac{B}{B-1} \sum_b m_b m_b'.$$

The excess is positive semidefinite. Estimated ratio denominators and centering require an asymptotic linearization, so this identity alone proves no finite-sample coverage. In a ten-village block with five fixed scores 4 and five 6 for every arm, assign one village each to lower and upper cash without replacement. Exhausting all 90 ordered pairs gives covariance $-1/(9B)$ between the two arm means across B independent blocks. The true difference variance is $20/(9B)$; discarding cross-arm dependence instead gives $2/B$, only 90 percent of that variance. The common block vector retains the missing covariance. The synthetic coverage failures in Table 3 show why repairing this algebra is insufficient to validate the subsequent multiplier approximation with 22 blocks.

Distributional priorities. On the finite integer support, any increasing function is its value at zero plus a nonnegative combination of survival indicators. Increasing-concave functions equivalently have nonnegative weights on truncated linear gains. Thus simultaneous distributional ordering would imply the corresponding class of preferences over this proxy. The empirical regions do not establish such an ordering. Preference over HDDS is distinct from preference over nutrients or full intertemporal resources.

B An illustrative saving mechanism

B.1 Household allocation

A household has current resources $w + t$, where w is its initial budget and t is a transfer. It allocates resources to current food expenditure f and saving s . Food is measured in units with price normalized to one. Let $\underline{f}$ denote subsistence expenditure, R the positive gross return on saving, $y > 0$ background future income, and $\beta > 0$ the weight on future consumption. For an

interior household with $w + t > \underline{f}$, preferences and the budget constraint are

$$U(f, s) = \log(f - \underline{f}) + \beta \log(Rs + y), \quad f + s = w + t, \quad s \geq 0. \quad (12)$$

The future resource y is important: it allows a change in the saving return to affect the allocation between dates. The model abstracts from risk, credit, household bargaining, and the multiple activities provided by Gikuriro. Its comparative statics describe a possible mechanism within that bundle.

Food diversity is an increasing step function of spending,

$$D(f) = \sum_{j=1}^{12} \mathbf{1}\{f \geq a_j\}, \quad \underline{f} \leq a_1 \leq \dots \leq a_{12}, \quad (13)$$

where a_j are household-specific thresholds for including additional food groups. This reduced form allows a household to maintain a staple diet while deferring more expensive sources of variety. It does not imply that each food group has the same nutrient content or that variety depends only on total expenditure.

Proposition 5 (Current food and saving). *If the optimum is interior, then*

$$f^* = \frac{w + t + y/R + \beta \underline{f}}{1 + \beta}, \quad s^* = \frac{\beta(w + t - \underline{f}) - y/R}{1 + \beta}.$$

Cash increases both current food expenditure and saving. A rise in R decreases current food expenditure, increases saving, and increases maximized household utility. Dietary diversity weakly increases with cash, but can remain unchanged over an interval of transfer values.

If saving is constrained at zero, a small change in its return has no effect on allocation; cash initially finances current spending. The saving-return prediction holds y fixed. I do not estimate R , y , or β : saving groups may change fees, credit, risk or commitments instead. The model illustrates how current food access and intertemporal utility can rank an intervention differently.

Corollary 3 (A short observation window). *A saving intervention can raise household utility while leaving the observed diversity score unchanged, or reducing it. A small cash transfer can raise food spending without increasing the diversity score. A larger transfer crosses a threshold only if an uncrossed threshold is reachable; a household already consuming all twelve groups cannot increase this score.*

The bundled assignment cannot distinguish this channel from delivery timing or other program components. Repeated measurements or randomized mechanism variation would be needed.

Proof of Proposition 5. Substitution of $s = w + t - f$ gives the first-order condition $1/(f - \underline{f}) = \beta R/(Rs + y)$. Strict concavity gives the unique interior optimum stated in the proposition. Its derivatives are $\partial f^*/\partial t = 1/(1 + \beta)$, $\partial s^*/\partial t = \beta/(1 + \beta)$, and $\partial f^*/\partial R = -y/\{(1 + \beta)R^2\} <$

0. The saving derivative has the opposite sign. The envelope derivative of maximized utility is $\beta s^*/(Rs^* + y) > 0$ at an interior positive-saving optimum. At the zero-saving boundary, the allocation response to a small return change is zero. Dietary diversity is a step function; transfer increases weakly raise it and may leave it unchanged. Once all twelve thresholds are crossed, the ceiling prevents a further score gain. No R , y , or β is calibrated from the bundled reduced form.

C Regression crosswalk and supplementary estimates

For comparison with the original study, the supplementary weighted ANCOVA is

$$v(D_{i1}) = \alpha + \sum_{a \neq 0} \tau_a T_{j(i),a} + \rho v(D_{i0}) + \gamma M_i + \lambda_{b(i)} + \varepsilon_i. \quad (14)$$

The baseline transformation is replaced by its weighted within-block mean where missing, with indicator M_i ; constant regressors are omitted. Source expansion weights enter WLS. CR1 uses $G/(G-1)(N-1)/(N-K)$, with pointwise t_{G-1} intervals. The original 204-test maximum-studentized family uses 9,999 village-score Rademacher draws; these are working asymptotic bands. Secondary outcomes have 96 distinct Holm tests, food indicators 80, baseline heterogeneity ten, and observation five. Every child and ineligible sensitivity has its own fifteen-test family. These separate families do not supply one paper-wide coverage claim.

The common-slope ANCOVA need not preserve assignment-adjusted target weights under heterogeneous effects. Complete-case selection remains unresolved. The coarse all-or-nothing bounds are retained in the replication tables solely as the initial sample crosswalk; the main item-informed conditional regions supersede their inferential interpretation. No statistical significance is required to retain an analysis, and every referee-driven extension is exploratory.

Table 10: Hypothetical value of unused expected resources

Groups per unused USD	Augmented-objective loss	Expected USD spent	Unused USD
0.000	7.720	80.75	43.73
0.001	7.700	80.67	43.81
0.005	7.634	78.70	45.79
0.010	7.551	71.80	52.69
0.020	7.208	58.70	65.78
0.050	5.267	25.79	98.70

Notes: Mean objective augmented by $\eta(b - c'q)$; the common comparator has the same resource value. Loss is in that augmented objective, not diet alone. Values of η are hypothetical and unestimated. Coherent finite-and-consistency region, original average costs and stable delivery.

Table 11: Source economic outcomes: exploratory assignment regressions

	Gikuriro	Lower	Middle	Upper	Large
Consumption (IHS)	−0.093 (0.099) [1.000]	0.118 (0.116) [1.000]	−0.001 (0.128) [1.000]	0.083 (0.120) [1.000]	0.327 (0.108) [0.245]
Productive assets (IHS)	0.008 (0.103) [1.000]	0.271 (0.127) [1.000]	0.248 (0.175) [1.000]	0.246 (0.135) [1.000]	0.780*** (0.118) [< 0.001]
Saving stock (IHS)	1.186* (0.338) [0.052]	−0.009 (0.447) [1.000]	−0.286 (0.539) [1.000]	0.353 (0.525) [1.000]	0.720 (0.322) [1.000]
Borrowing stock (IHS)	0.056 (0.358) [1.000]	−0.667 (0.339) [1.000]	−0.714 (0.434) [1.000]	−0.830 (0.375) [1.000]	−0.325 (0.388) [1.000]
Health knowledge index	−0.062 (0.366) [1.000]	0.273 (0.488) [1.000]	0.041 (0.410) [1.000]	0.068 (0.373) [1.000]	0.039 (0.474) [1.000]
Sanitation practices index	−0.207 (0.227) [1.000]	0.337 (0.363) [1.000]	−0.229 (0.286) [1.000]	0.323 (0.283) [1.000]	0.112 (0.298) [1.000]
Purchased food (IHS)	−0.172 (0.177) [1.000]	0.058 (0.192) [1.000]	0.024 (0.187) [1.000]	0.264 (0.188) [1.000]	0.267 (0.170) [1.000]
Own-produced food (IHS)	0.125 (0.322) [1.000]	0.556 (0.378) [1.000]	−0.601 (0.427) [1.000]	−0.558 (0.556) [1.000]	0.968 (0.393) [1.000]
Households (range across outcomes)	1751	1751	1751	1751	1751
Villages (range across outcomes)	248	248	248	248	248

Notes: IHS coefficients are transformation units, not percentage changes. Source preprocessing is inherited except the stated food-value IHS transformation. Village CR1 SEs and the separate 96-test economic Holm family. No causal mediation or total-welfare interpretation.

Table 12: Follow-up child scores in the fixed baseline-flagged cohort

Outcome	Gikuriro	Lower	Middle	Upper	Large
Height-for-age	0.042 (0.050)	0.031 (0.062)	-0.080 (0.077)	-0.038 (0.061)	0.090 (0.056)
Holm p	1.000	1.000	1.000	1.000	1.000
N	2141	2141	2141	2141	2141
Weight-for-age	0.042 (0.040)	0.008 (0.056)	-0.019 (0.057)	0.041 (0.043)	0.081 (0.039)
Holm p	1.000	1.000	1.000	1.000	0.566
N	2131	2131	2131	2131	2131
Arm circumference	0.024 (0.056)	-0.029 (0.078)	0.009 (0.107)	-0.070 (0.067)	0.140 (0.078)
Holm p	1.000	1.000	1.000	1.000	1.000
N	1837	1837	1837	1837	1837

Notes: Membership fixed by baseline eligibility and source anthropometry flag, regardless of endline eligibility/due status. Regression uses available follow-up scores, original assignment/weights, baseline source score, missing-lag indicator, and blocks. No full-cohort selection-robust effect is claimed. Village CR1 SEs; fifteen-test Holm family separate from the endline-due analysis.

Table 13: Food-module information in the baseline eligible cohort

Arm	Baseline	Identified	Partial	Unobserved
Control	521	498	6	17
Gikuriro	540	522	2	16
Lower	165	160	0	5
Middle	154	149	0	5
Upper	167	164	3	0
Large	246	237	4	5

Notes: Identified includes complete modules and partial modules that nevertheless identify all twelve groups. Partial denotes an interval of width between one and eleven. Unobserved denotes width twelve, including absent endline households. Every baseline eligible household occurs once. These counts are not a missing-at-random adjustment.

Table 14: Crosswalk from the original diet specification

Specification	GK	GK SE	Large	Large SE	N
Original controls, pooled cash	0.192	0.122	0.550	0.126	1751
Original controls, split cash	0.201	0.123	0.558	0.126	1751
Supplied-score lag only	0.235	0.131	0.530	0.131	1751
Integer outcome, supplied lag	0.258	0.130	0.547	0.133	1728
Integer outcome and baseline	0.255	0.131	0.545	0.134	1728

Notes: Original controls reproduce the fixed covariate list in the corrected deposit's `CovariateLists.xlsx` and `Table_3.do`, including the supplied lag. Pooled means the three smaller cash arms share a coefficient. Split estimates them separately. Collinear supplied baseline columns are omitted in the stated order, matching the identified regression span. Later rows change the control list and integer-score missingness sequentially. Standard errors use village CR1. The final row is this project's original ANCOVA. Published rounding, rather than an exact recovery of a withdrawn dataset, is the comparison.

Table 15: Available child growth and ineligible-household outcomes

Outcome	GK	SE	Holm p	Large	SE	Holm p	N
Child height-for-age	0.004	0.057	1.000	0.086	0.069	1.000	2607
Child weight-for-age	-0.003	0.040	1.000	0.092	0.045	0.608	2593
Child arm circumference	-0.023	0.053	1.000	0.148	0.073	0.616	2162
Ineligible HDDS	0.111	0.142	1.000	-0.392	0.209	0.798	960
Ineligible consumption (IHS)	-0.149	0.118	1.000	-0.154	0.201	1.000	967
Ineligible saving (IHS)	-0.334	0.423	1.000	-0.931	0.455	0.629	967

Notes: Source-weighted assignment effects with block/baseline controls and village-clustered SE. Child standardized scores use the source endline due-for-measurement sample; ineligible households use baseline eligibility. Separate fifteen-test Holm families include all five arms; counts differ by outcome. These selected exploratory samples do not replace the parent paper's richer prespecified anthropometric regressions.

Table 16: Assignment effects on observing an endline diet

Arm	Observed-diet effect	SE	Holm p
Gikuriro	1.29	1.04	0.875
Lower	0.32	1.60	1.000
Middle	1.48	1.47	0.953
Upper	3.11	1.10	0.026
Large	0.72	1.27	1.000

Notes: Effects and standard errors are in percentage points. The outcome is one if a baseline eligible household is in the endline panel with a complete food module, and zero otherwise. Estimates use baseline sampling weights, block fixed effects and village-clustered standard errors. Holm adjustment covers the five arm-versus-control observation tests. This is a distinct diagnostic family; no observation effect proves ignorability of missing diet.

D Established-method comparisons and component ablations

This appendix assesses which implementation choices change the empirical guarantee while preserving the same weighted released-baseline target. Averaging expectation-bounded nonnegative tests is established e-value theory (Vovk and Wang, 2021); the shared aggregation is not a new theorem. Range comparisons use Bardenet and Maillard (2015), including the harmonic sum in the proof of Proposition 2.2. I also implement a weight-sensitive product bound and a restricted empirical-Bernstein adaptation of Waudby-Smith and Ramdas (2020). The latter uses fixed forecasts and baseline-selected scales. It is not their complete adaptive-scale procedure, and the comparisons do not establish superiority over modern betting methods.

For one block write $t_j = W_j(y_j - u)$, $W_{\max} = \max_j W_j$, and $G = (n/k) \sum_{j \in A} t_j - \sum_j t_j$. Since $0 \leq y_j, u \leq 1$, the t_j have range at most $W_{\max}$. Under uniform quota sampling, the standard exponential inequalities imply $\log \mathbb{E} e^{\lambda G} \leq \lambda^2 V/8$, with forward coefficients

$$\frac{V_H}{W_{\max}^2} = \frac{n^2}{k}, \quad \frac{V_S}{W_{\max}^2} = \frac{n^2}{k} \left(1 - \frac{k-1}{n}\right), \quad \frac{V_M}{W_{\max}^2} = \frac{n^2(n-k)^2}{k^2} \sum_{t=1}^k \frac{1}{(n-t)^2}.$$

The harmonic expression applies for $k < n$; at $k = n$ the residual is zero. Complementing the sampled subset gives $G_k = -(n-k)G_{n-k}/k$, so I use the smaller forward or rescaled complementary coefficient. Selection uses baseline design quantities alone. Independent blocks add their coefficients.

For completeness, choosing predictable scales $\lambda_t = \Lambda(n-k)/(n-t)$ in the without-replacement Hoeffding process, with $\Lambda = \lambda n/k$, makes the coefficient on sample position j equal to

$$\lambda_j + \sum_{t=j+1}^k \frac{\lambda_t}{n-t+1} = \Lambda.$$

Its centered outcome sum is therefore λG ; its quadratic compensation is the displayed harmonic coefficient. This retains an established martingale calculation before the integral relaxation leading to Serfling’s simpler bound. It is not a new concentration inequality.

Every field alternative uses the same item intervals, 23 normalized score functions, deterministic restrictions, costs and rational proposals. Its 276 expectation-bounded terms yield a standalone conditional 95 percent shared event. Classical scales minimize their quadratic radius analytically; product and hybrid scales use a baseline-only scalar search; the quota procedure retains its four-point baseline grid. The empirical-Bernstein procedure instead compensates the realized village scores and uses the fixed population denominator. Support scales may depend on outcomes because they certify deterministic bounds within an event. No observed minimum across separately valid events is asserted as a jointly valid 95 percent guarantee.

The shared sum constraint implies every individual tail inequality, so the quota shared region is contained in the separable Bonferroni region when constants and floors are held fixed. Differences between valid upper certificates need not equal differences in the exact support or optimized loss.

Table 17: Existing-method comparisons for the same allocation proposals

Moment/event	Mean loss		Shortfall loss	
	Lower	Upper	Lower	Upper
Quota, shared event	5.866	5.868	0.502	0.503
Quota, individual tests	—	6.878	—	0.577
Quota, shared, no bin floors	5.865	5.868	0.502	0.503
Hoeffding, shared	6.639	6.959	0.560	0.571
Serfling, shared	6.546	6.632	0.547	0.558
Harmonic martingale, shared	6.541	6.616	0.546	0.558
Weight-sensitive product, shared	7.612	7.815	0.614	0.642
Product/harmonic hybrid, shared	6.532	6.609	0.546	0.557
Empirical Bernstein, fixed forecast	7.575	7.812	0.608	0.645

Notes: Lower regional certificates are rounded downward, and upper proposal certificates upward, to three decimals. The separate-event row has no regional lower bound here. Mean and shortfall columns use their respective fixed rational proposals and original budget/comparator menu. Statistical scales use baseline quantities; the empirical-Bernstein compensation uses realized score intervals and averages all assigned-village orders. Support multipliers are selected numerically and independently enclosed. Each row has its own conditional event; joint coverage across rows is not asserted. The lower/upper pairs bracket method-specific optima only within their implemented outer regions. Separate quota tests retain the same statistical scales and constants; removing bin floors retains scalar logical consistency.

Nearly identical displayed mean bounds with and without bin floors do not support attributing substantial mean tightening to that component.

The inexpensive weight-sensitive comparator substantially changes the apparent gain. At the half quota with one heavy village, the old six-point quota radius is 0.394 versus a harmonic radius of 0.777, but the product radius is 0.419. The comparable baseline searches give 0.391 versus 0.419: a 6.74 percent reduction, rather than the 49.23 percent reduction against the range envelope. Across all 33 cases the common-search quota improvement over the hybrid lies between 1.39 and 6.74 percent. These are gains in a specified relaxation, not uniform advantages for treatment decisions.

D.1 Certified regional optimization brackets

The displayed lower bounds concern the implemented outer sets defined by the stored outward affine tests, logical rows and bin floors. For each method I construct 19 exact-rational probability populations, including an interior seed and candidates informed by the existing support searches. Unit mass, nonnegativity, floors and logical inequalities are checked with exact fractions. A directed exponential sum no greater than 5520 certifies membership in the implemented event. Floating searches choose displacements toward the seed; only the final directed membership check admits a population. No numerical feasibility tolerance admits one.

Let μ^m be its exact arm-mean vector and let each index ℓ pair a feasible population with a budget-feasible comparator vertex r_ℓ . Put $a_\ell = r'_\ell \mu^m$. Any nonnegative probabilities w_ℓ summing

Table 18: Weight concentration and quota-moment gains in twelve-village blocks

Quota	Weights	Quota radius	Hybrid radius	Reduction (%)
1	Equal	0.432	0.443	2.31
1	Alternating	0.648	0.664	2.31
1	Concentrated	1.862	1.913	2.67
3	Equal	0.229	0.232	1.39
3	Alternating	0.344	0.348	1.39
3	Concentrated	0.802	0.838	4.27
6	Equal	0.133	0.136	2.35
6	Alternating	0.199	0.204	2.35
6	Concentrated	0.391	0.419	6.74
11	Equal	0.039	0.040	2.31
11	Alternating	0.059	0.060	2.31
11	Concentrated	0.169	0.174	2.67

Notes: Twenty-two identical independent blocks, family penalty $\log 5520$, normalized scalar tail radius. Equal weights are all one; alternating weights are one/three; concentrated weights are eleven ones and one ten. Radii above one remain untruncated. Both columns use the same baseline-only search bounds and six initial scales, followed by enclosed moment and division calculations. The hybrid takes the smaller product/harmonic bound on each moment before any outcomes are observed. Reductions compare valid upper radii, not certified globally optimized radii. The complete catalogue retains its original six- and ten-village cases and historical grid results. These are designed examples, not field outcomes, allocation losses or coverage estimates.

to one and any $\eta \geq 0$ give, for every $q \in \mathcal{Q}(b)$,

$$\max_{\ell} \{a_{\ell} - (\mu^m)'q\} \geq \sum_{\ell} w_{\ell} a_{\ell} - \eta b - \max_j \left\{ \sum_{\ell} w_{\ell} \mu_j^m - \eta c_j \right\}.$$

The first inequality averages the finite adversary’s losses; the second uses $c'q \leq b$, nonnegative shares and unit mass. I extract candidate probabilities and a cost multiplier from a finite policy LP, normalize and clamp them to be feasible, and evaluate this expression exactly. Its nonnegative part is a regional minimax lower bound. A separately verified feasible proposal supplies the upper bound. Neither a floating primal value nor a purported optimizer gap supplies either certificate.

The quota mean bracket is $[5.866228, 5.867129]$, with certified width 0.000901; the shortfall bracket is $[0.502242, 0.502549]$, with width 0.000306. The hybrid’s regional mean lower bound exceeds 6.532 groups, so even its optimized allocation cannot close the difference with this quota proposal within these particular outer sets. The lower distributions are not claimed to be jointly realizable with the observed finite individual weights, and membership in the stored outer event is not membership in a sharper conceptual or attainable-population set. These brackets explain where optimization matters in the implemented comparison; they do not establish statistical minimax risk, impossibility of sharper inference, or a jointly protected choice among methods.

D.2 Computing the relaxed quota moment

For $u = 0$, write $x_j = W_j y_j \in [0, W_j]$ and

$$F(x) = \binom{n}{k}^{-1} \sum_{|A|=k} \exp \left\{ \lambda \left(\frac{n}{k} \sum_{j \in A} x_j - \sum_j x_j \right) \right\}.$$

This is symmetric and convex. Fix a maximizing binary vertex with m active coordinates. If an active weight W_i is below an inactive weight W_j , remove the active coordinate and regard its value as a scalar argument s of the resulting symmetric convex function $g(s)$. Global maximality gives $g(W_i) \geq g(0)$. Convex secant slopes imply $g(W_j) \geq g(W_i)$, so exchanging the active indices preserves a maximum. Repeating exchanges leaves the m largest weights active. The same argument holds for the negative sign; $u = 1$ reduces to that sign by complementation. Thus $2(n + 1)$ sorted populations suffice. General symmetric-convex range computation under interval uncertainty predates this application (Xiang, 2006). The argument here supplies a self-contained specialization, not a claim to a new general algorithm.

For a sorted population with total S , put $a_j = \exp\{s\lambda(nx_j - S)/k\}$, where $s \in \{-1, 1\}$. Its quota average is the elementary symmetric mean, computed by

$$D_{t,r} = \frac{t-r}{t} D_{t-1,r} + \frac{r}{t} a_t D_{t-1,r-1}, \quad D_{t,0} = 1.$$

The terminal value $D_{n,k}$ is the average of the products over all k -subsets. Complementation permits evaluation at $q = \min(k, n - k)$; the exponent denominator remains the original k . Directed argument, exponential, positive-coefficient recurrence and logarithm evaluations enclose the maximum. The arithmetic cost is $O(n^2 q)$ with $O(n + q)$ working storage. An independent 160-digit full population/subset calculation checks the implementation on small fixtures; the exhaustive proxy program remains available.

The declared computation catalogue contains 16 designs, with block sizes 12, 24, 48 and 96, quarter/half quotas, and equal/concentrated weights. It times three actual serial calls per method at a fixed scale. Exhaustive computation is skipped when $2^n \binom{n}{k}$ exceeds four million, with no extrapolated timing claim. The 96-village half-quota calculations take roughly two seconds on the recorded Windows interpreter; the inexpensive hybrid remains substantially faster. Timings in the machine-readable receipt concern moments, not end-to-end allocation optimization.

D.3 Weight-sensitive product and empirical compensation

Let $p = k/n$ and $I_j = \mathbf{1}\{j \in A\}$. For nonnegative a_j , expanding the product and using $\Pr(S \subset A) = \binom{k}{|S|} / \binom{n}{|S|} \leq p^{|S|}$ gives $\mathbb{E} \prod_j (1 + a_j I_j) \leq \prod_j (1 + p a_j)$. Complementary inclusion indicators give the negative-sign analogue. This is the familiar negative-association comparison (Joag-Dev and

Proschan, 1983). Convex endpoint reduction and monotonicity in nonnegative amplitudes yield

$$B_P = \max_{s \in \{-1, 1\}} \sum_j \log \left\{ (1-p)e^{-s\lambda W_j} + pe^{s\lambda(n-k)W_j/k} \right\}.$$

The minimum of B_P and the harmonic quadratic bound still bounds the same block moment. Because it is fixed from baseline design quantities, using that minimum does not select or intersect observed confidence events. At a half quota the product simplifies to $\sum_j \log \cosh(\lambda W_j)$.

For the empirical-Bernstein comparison, normalize $X_j = W_j y_j / W_{\max} \in [0, 1]$ and fix a forecast $\gamma = \sum_j W_j / (2nW_{\max})$. Use predictable scales $\theta_t = \lambda(n/k)W_{\max}(n-k)/(n-t) < 1$ and compensation $\rho(\theta) = -\log(1-\theta) - \theta$. The published supermartingale argument (Waudby-Smith and Ramdas, 2020) permits a predictable bounded forecast. The telescoping coefficients give

$$M_s = \exp \left\{ s\lambda G(y, 0) - \sum_{t=1}^k \rho(\theta_t) (X_{A_t} - \gamma)^2 \right\}, \quad \mathbb{E} M_s \leq 1.$$

For either sign, each observed contribution is concave in X_{A_t} , so its minimum over the item interval occurs at an endpoint. Directed endpoint minima give a pathwise lower test without an ignorability condition. The field implementation averages the positive tests over all $k!$ orders of the observed villages; it does not presume that village identifiers were randomly ordered. Independent blocks multiply these averages. Their population factors cancel at the full weighted mean, leaving affine observable log tests with coefficient $\pm\lambda N$, where N is fixed total weight. The 276 tests therefore have the same Markov failure bound of 0.05. The baseline scale heuristic assumes a uniform unit score only to choose an admissible scale; actual compensation uses the observed endpoints. Fixed forecasts, the common scale, order averaging and conservative endpoint minimization limit this adaptation. Its field bounds, 7.812 and 0.645, cannot be taken as evidence against the full variance-adaptive or betting family.

D.4 Known-population decision performance

To distinguish a moment gain from a decision gain, I predefine 144 populations: 8, 24 or 64 villages per block; 8 or 32 blocks; equal weights or one weight ten; effects -0.16 , zero, 0.04 or 0.16 ; and complete scores, intervals of width 0.2 , or reporting that depends on assigned arm and its potential score. Scores and weights are fixed, with the full weighted population mean known. Each cell has 256 half-quota draws at seed 20261001, with a separate uniform ordering for each arm. The four original methods receive exactly the same assignment draws as before adding empirical Bernstein. Every cell remains reported.

This experiment has two equally costly arms and one normalized mean, with four Bonferroni exponential tests at cap 80. It is a different procedure from the field distributional event. For its rectangular mean region, put $U = \max(0, H_1 - L_0)$ and $L = \max(0, H_0 - L_1)$. Assigning arm 1 with probability q has worst-case loss $\max\{(1-q)U, qL\}$, minimized at $q = U/(U+L)$ with value $UL/(U+L)$ when $U+L > 0$. Exact rational arithmetic verifies the lower value and

the upper loss after rounding q to millionths; the error is at most $\max(U, L)/(2 \cdot 10^6)$. Empty statistical intersections fall back to the deterministic logical region and their counts remain visible. The empirical-Bernstein endpoint contributions are rounded downward to a specified dyadic grid; interval and policy arithmetic then remain exact.

Table 19: Known-population decision performance across the declared grid

Reporting	Method	Actual regret	Certificate	At most 0.10 (%)
Complete	Quota	0.019	0.114	50.1
Complete	Harmonic	0.025	0.156	35.7
Complete	Product	0.026	0.158	25.6
Complete	Hybrid	0.020	0.123	44.5
Complete	Empirical Bernstein	0.017	0.101	53.7
Coarse interval	Quota	0.031	0.210	4.3
Coarse interval	Harmonic	0.033	0.235	4.3
Coarse interval	Product	0.034	0.247	0.0
Coarse interval	Hybrid	0.032	0.218	4.3
Coarse interval	Empirical Bernstein	0.029	0.195	12.6
Assignment-dependent	Quota	0.039	0.320	0.0
Assignment-dependent	Harmonic	0.040	0.332	0.0
Assignment-dependent	Product	0.040	0.343	0.0
Assignment-dependent	Hybrid	0.039	0.324	0.0
Assignment-dependent	Empirical Bernstein	0.039	0.318	0.0

Notes: Equal-weight averages over all 48 cells per reporting pattern, with 256 draws per cell. Actual regret uses known potential means; the certificate is an enclosed upper loss for the stated rectangular region. The last column is the fraction of draws with a certificate at most 0.10 normalized units. These are designed populations, not Rwanda estimates, normative organizational tolerances, or proof of coverage. All 720 cell/method rows, rounding bounds, computation times, coverage and fallback counts are available.

With complete scores, average actual regret is 0.019 for quota and 0.020 for the cheap hybrid, while the fixed-forecast empirical-Bernstein rule gives 0.017. Its average certificate is also smaller. Coarse reporting reverses neither conclusion, and assignment-dependent nonreporting leaves all methods with broad certificates and no certificates at or below 0.10. The realized interval coverage is one in every cell, indicating conservative performance on these populations; it was never used as a pass threshold or to retune the methods. These calculations rule out a broad dominance interpretation of the quota construction. They do not change the empirical conclusion that the Rwanda evidence leaves substantial decision uncertainty.

E Replication and verification

The public package contains a single master script, frozen independent input hashes, corrected source attribution, codebooks, extraction programs, dependency locks, numerical outputs, generated LaTeX tables and in-text macros, and the compiled PDF. The source ZIP is excluded; its numeric research extract is redistributed under CC BY 4.0. Recreating this extract from the verified archive is distinct from recreating the parent’s entire upstream survey processing.

The initial mean ANCOVA is checked by an independent Stata implementation. Scientific in-

variants, policy-vertex comparisons against independent linear programs, bounded-sampling exponential moments, block-score algebra, sparse-outcome support, cohort partitions, and primal/dual adversary values are checked separately. The master retains failed synthetic coverage results; a successful implementation check cannot turn them into passed coverage tests. Cold-output and separate-environment comparisons record their actual source version and scope in the public audit. They do not imply independent data-editor replication.

The JPE-template report is a self-audit, rather than an official journal report. Genuine ChatGPT referee text and point-by-point responses are preserved separately. Simulated referee recommendations are not real journal acceptance. The original secondary plan and all amendments remain available, including the replacement of the unsupported village-score certificates by the stated finite conditional outer regions and exploratory block sensitivity.